

Lecture 4 - Kinetic energy, work, potential energy, and power

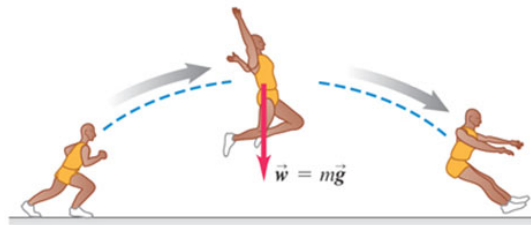
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Kinetic energy is due to speed. Potential energy is due to configuration (how it looks like, e.g., look like very high.)

In this lecture we will consider how these 2 types of energy are related to each other.

Power (or powerfulness!!) can be related to how desperate you work. If you have done a lot of work, are you more powerful? We will discuss.



(Powerfulness is a simple word used by A/Prof Tay to explain Power. In Physics the concept is called Power and there isn't such a word called "powerfulness" in the Physics textbooks.)

A few slides in this chapter come from Chapter-6 and Chapter-7 of Young's textbook.

This PPT file will well define the scope of your tests.

1

Kinetic energy and work

- A particle of mass m moving at **speed** v has

kinetic energy KE , where $KE = \frac{1}{2}mv^2$.

(Translational)
Kinetic Energy

The work done by the **net** force on an object will be exhibited in the change of velocity for the object.

- If the particle is subject to an external **net force**, the changes in **kinetic energy** for the particle is ΔKE , where

$$\Delta KE = KE_{\text{final}} - KE_{\text{initial}} = W_{\text{done by force}}$$

and W is the **work**, and is equal to the force x displacement along the same direction.

Work-energy theorem.

- Unit of kinetic energy and work (same unit because work is also energy!!)

$$\begin{aligned} 1 \text{ joule (J)} &= 1 \text{ kg} \times (1 \text{ m/s})^2 \\ &= (1 \text{ kg} \times \text{m/s}^2) \times 1 \text{ m} = 1 \text{ N} \times 1 \text{ m}. \end{aligned}$$

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Example 1. Gravitational potential energy – a freely falling object

- Mass of the object is m
- At y_1 , **kinetic energy** of the object,

$$KE_1 = \frac{1}{2}mv_1^2.$$

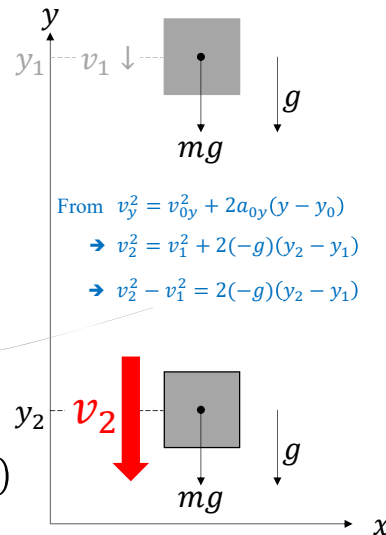
- At y_2 , **kinetic energy** of the object,

$$KE_2 = \frac{1}{2}mv_2^2.$$

- The **change in kinetic energy** of the object is

$$\begin{aligned}\Delta KE &= KE_2 - KE_1 = \frac{1}{2}m(v_2^2 - v_1^2) \\ &= \frac{1}{2}m(2(-g)(y_2 - y_1)) \\ &= -(mg(y_2 - y_1))\end{aligned}$$

$$\rightarrow \Delta KE = -\Delta PE \quad PE : \text{Potential energy}$$



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Example 1. Gravitational potential energy – a freely falling object (cont.)

- **Gravitational potential energy** = $mg \times y$
where y is measured from the **zero potential level**
- The **change in kinetic energy (ΔKE)** of the object is

$$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = -(mgy_2 - mgy_1).$$

- The **change in gravitational potential energy (ΔPE)** is $\Delta PE = mgy_2 - mgy_1$.

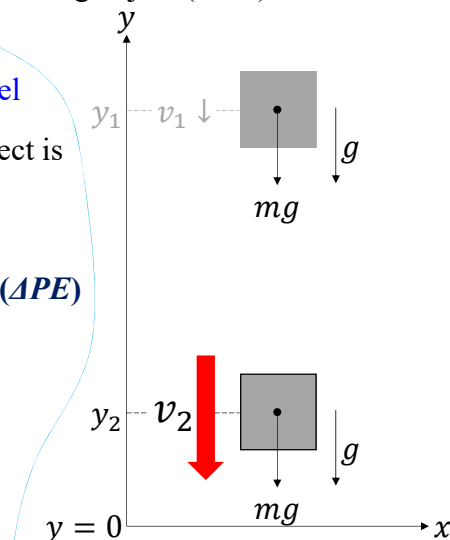
- For a freely falling object, the change in kinetic energy,

$$\Delta KE = W_{\text{done by gravity}} = -\Delta PE.$$

Gain in KE is at the expense of the loss in PE.

Gain in PE is also at the expense of the loss in KE.

What if the acceleration of the object is a , and $a \neq g$?



Answer: There must be some external forces such as the air friction !!!

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Problem. The mechanical system shown in the figure consists of a light inextensible cord, light frictionless pulleys, and blocks of equal mass. It is initially held at rest so that the blocks are at the same height above the ground. The blocks are then released. Find the speed of block A at the moment when the vertical separation of the blocks is h .

Answer:

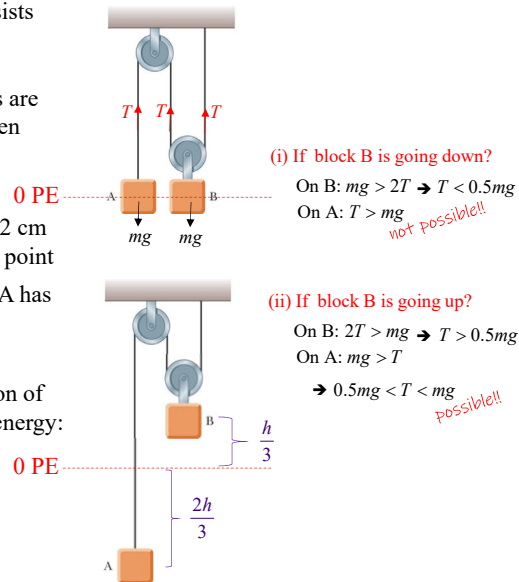
If block B moves up by 1 cm, block A will move down by 2 cm and their separation becomes 3 cm. So we choose the final point to be when B has moved up by $\frac{h}{3}$ and has speed v_B . Then A has moved down $2 \times \left(\frac{h}{3}\right)$ and has speed $2v_B$.

Take the initial position as 0 potential level. By conservation of mechanical energy, gain in energy comes from the loss in energy:

$$\frac{1}{2}m(2v_B)^2 + \frac{1}{2}m(v_B)^2 + mg\left(\frac{h}{3}\right) = mg\left(\frac{2h}{3}\right)$$

$$\frac{5}{2}v_B^2 = \frac{gh}{3} \Rightarrow v_B = \sqrt{\frac{2gh}{15}}$$

$$v_A = 2v_B = 2 \times \sqrt{\frac{2gh}{15}} = \sqrt{\frac{8gh}{15}}$$



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Example 1. Gravitational potential energy – a freely falling object (cont.)

- The position and velocity of a **freely falling object** change in the course of time in such a way that

$$\Delta KE + \Delta PE = 0.$$

In isolated system, the total mechanical energy is conserved.

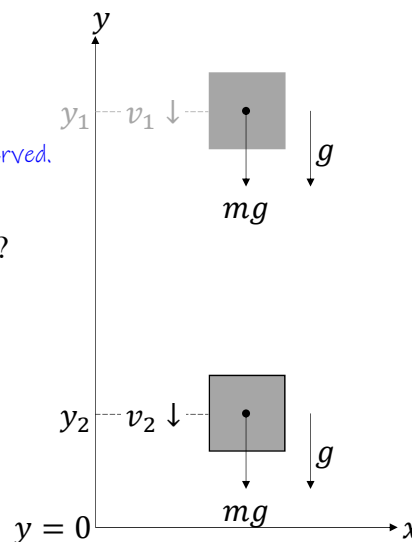
Questions.

- How about an object in **projectile motion**?

*Isolated system
When a projectile is higher, its speed must be slower, and vice versa.*

- What happens when an additional force acts to the weight of the projectile gives $\Delta KE + \Delta PE \neq 0$?

That force must be non-conservative, such as the air resistance or friction.



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Conservative and Nonconservative Forces

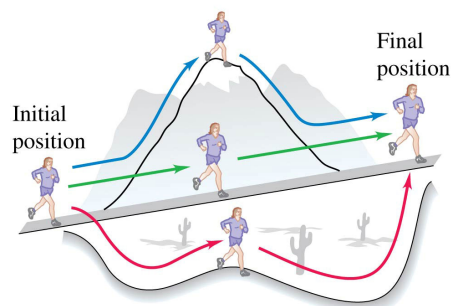
- A **conservative force** allows conversion between kinetic energy and potential energy. Gravity and the spring force are conservative.
- The work done between two points by any conservative force
 - a) can be expressed in terms of a **potential energy function**.
 - b) is reversible.
 - c) is independent of the path between the two points.
 - d) is zero if the starting and ending points are the same.
- A force (such as friction) that is not conservative is called a **nonconservative force**, or a **dissipative force**.

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Conservative Forces

- The work done by a conservative force such as gravity depends on only the endpoints of a path, not the specific path taken between those points.

Because the gravitational force is conservative, the work it does is the same for all three paths.



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Problem on Conservative Forces. A skier starts at the top of a very large, frictionless snowball, with a very small initial speed, and skis straight down the side as shown in figure. At what point does she lose contact with the snowball and fly off at a tangent, i.e., find the angle for α ? Treat the skier as a point mass.

Answer:

Let n be the normal force on the skier, v be the tangential velocity and r be the radius of the big snow ball. The vertical circular motion at the fly-off point is:

$$mg \cos \alpha - n = m \left(\frac{v^2}{r} \right) \dots (1)$$

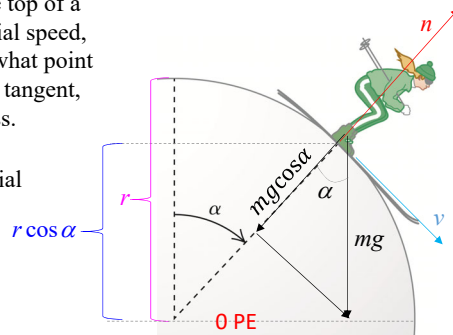
By conservation of energy: $mgr = \frac{1}{2}mv^2 + mg(r \cos \alpha) \dots (2)$

$$\text{Sub } v^2 \text{ in (1) to (2): } mgr = \frac{1}{2}m(rg \cos \alpha) + mg(r \cos \alpha) \rightarrow 1 = \frac{1}{2} \cos \alpha + \cos \alpha = \frac{3}{2} \cos \alpha$$

$$\rightarrow \alpha = \cos^{-1} \frac{2}{3} = 48.19^\circ$$

What we have learnt: As the skier loses gravitational potential energy, she gains kinetic energy so she speeds up. She loses contact when she is going so fast that the radially inward component of her weight isn't large enough to keep her in the vertical circular path. Note that where she loses contact does not depend on her mass or on the radius of the snowball.

But we still have not qualitatively explain the reason for the skier to leave the surface!!



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Nonconservative Forces

- As an automobile tire flexes as it rolls, nonconservative internal friction forces act within the rubber.
- Mechanical energy is lost and converted to internal energy of the tire.
- This causes the temperature and pressure of a tire to increase as it rolls.
- That's why tire pressure is best checked **before** the car is driven, when the tire is **cold**.



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Conservation of Energy

- Nonconservative forces do not store potential energy, but they do change the **internal energy** of a system.
- **The law of conservation of energy** means that energy is never created or destroyed; it only changes form.
- This law can be expressed as

$$\Delta K + \Delta U + \Delta U_{\text{int}} = 0.$$

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Example on Nonconservative Forces. 200 kg steel hammerhead of a pile driver is lifted 3.00 m above the top of a vertical I-beam being driven into the ground. The hammerhead is then dropped, driving the I-beam 7.40 cm deeper into the ground. The **vertical guide rails exert a constant 60.0 N friction force** on the hammerhead during its descent in the air and the penetration into the ground. Ignore the effects of the air.

(i) Find the speed v of the hammerhead just as it hits the I-beam.

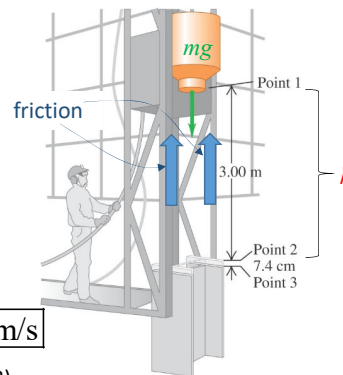
Answer:

Total mechanical energy is **not conserved** so we cannot use $\frac{1}{2}mv^2 = mgh$.

Loss in mechanical energy will be given to (is due to) the work done by friction on the guide rails.

$$mgh = \frac{1}{2}mv^2 + \text{friction} \times \text{displacement } h$$

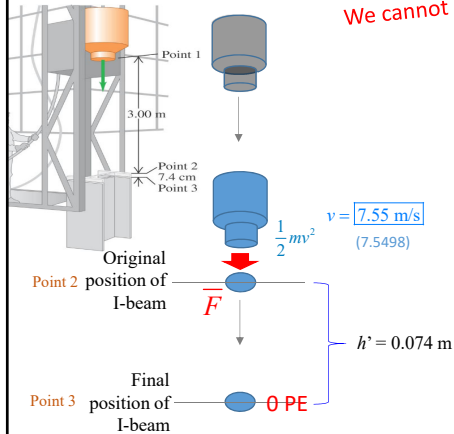
$$200 \times 9.8 \times 3 = \frac{1}{2} \times 200 \times v^2 + 60 \times 3 \rightarrow v = \boxed{7.55 \text{ m/s}} \\ (7.5498)$$



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- (ii) Find the average force the hammerhead exerts on the I-beam.

Answer:



We cannot use $\left(\frac{1}{2}mv^2 + mgh'\right) = F \times h'$!!!

Example on Nonconservative Forces. 200 kg steel hammerhead of a pile driver is lifted 3.00 m above the top of a vertical I-beam being driven into the ground. The hammerhead is then dropped, driving the I-beam 7.40 cm deeper into the ground. The vertical guide rails exert a constant 60.0 N friction force on the hammerhead during its descent in the air and the penetration into the ground. Ignore the effects of the air.

The work used to drive the I-beam into the ground comes from the mechanical energy of the hammerhead but there is a leakage again due to the friction (60 N) on the guide rails for the subsequent 0.074 m.

After the I-beam is driven 0.074 m deeper into the ground, we take this final position as 0 potential level. Let $\bar{F}$ be the average force the hammerhead exerts on the I-beam.

$$\left(\frac{1}{2}mv^2 + mgh'\right) = \begin{array}{l} \text{Energy loss on} \\ \text{guide rails due to} \\ \text{friction} \end{array} + \underbrace{\bar{F} \times h'}_{\text{work done by hammerhead}}$$

$$\rightarrow \left(\frac{1}{2} \times 200 \times 7.5498^2 + 200 \times 9.8 \times 0.074\right) = \text{friction} \times h' + \bar{F} \times 0.074$$

$$\rightarrow \left(\frac{1}{2} \times 200 \times 7.5498^2 + 200 \times 9.8 \times 0.074\right) = 60 \times 0.074 + \bar{F} \times 0.074 \rightarrow \bar{F} = \boxed{78926.32 \text{ N}}$$

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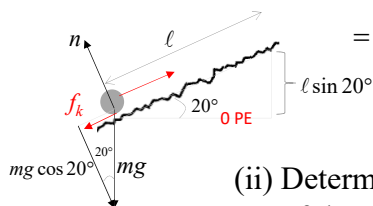
Problem. A crate of mass 10.0 kg is pulled up a rough incline with an initial speed of 1.50 m/s. The pulling force is 100 N parallel to the incline, which makes an angle of 20.0° with the horizontal. The coefficient of kinetic friction is 0.400, and the crate is pulled 5.00 m along the inclined surface. Treat the base of incline as zero potential level.

- (i) How much gravitational potential energy is gained (or lost) by the crate?

Answer:

$$\Delta PE = (mg) \times \Delta h = mg \times (\ell \sin \theta)$$

$$= (10.0 \text{ kg})(9.80 \text{ m/s}^2)(5.00 \text{ m}) \sin 20^\circ = \boxed{167.60 \text{ J}}$$



(The crate **gains** gravitational potential energy.)

- (ii) Determine the increase in internal energy of the crate-incline system due to friction.

Answer:

$$\Delta E_{\text{int}} = f_k \times \ell = (\mu_k n) \times \ell = (\mu_k (mg \cos \theta)) \times \ell$$

$$= ((0.400)(10.0)(9.80) \cos 20.0^\circ) \times 5.00 \text{ m} = \boxed{184.18 \text{ J}}$$

due to increase in temperature as the friction rubs the rough surface

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(iii) How much work is done by the **100-N** force on the crate?

Answer:

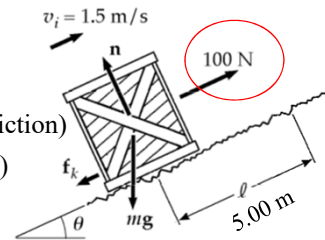
$$W_F = F \times \ell_{\text{along the same direction}} = (100)(5.00) = \boxed{500 \text{ J}}$$

(iv) What is the change in kinetic energy of the crate?

Answer:

$$\Delta KE = (\text{Supplied Energy by external force}) - (\text{Energy wasted by Friction}) \\ - (\text{Potential Energy gained by Crate})$$

$$= 500 - 184.16 - 167.60 = \boxed{148.24 \text{ J}}$$



(v) What is the speed of the crate after being pulled 5.00 m?

Answer:

$$\Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \rightarrow \frac{2(\Delta KE)}{m} = v_f^2 - v_i^2$$

$$\rightarrow v_f = \sqrt{\frac{2(\Delta KE)}{m} + v_i^2} = \sqrt{\frac{2(148.24)}{10.0} + (1.50)^2} = \boxed{5.65 \text{ m/s}}$$

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Example 1. Gravitational potential energy – a freely falling object (cont.)

Recall Lecture 2, Example 1. For an object of mass m in **projectile motion**,

$$x(t) = x_0 + v_{0x}t, \quad y(t) = y_0 + v_{0y}t - \frac{1}{2}gt^2.$$

$$\text{So, } v_x(t) = \frac{d}{dt}x(t) = v_{0x}, \quad v_y(t) = \frac{d}{dt}y(t) = v_{0y} - gt.$$

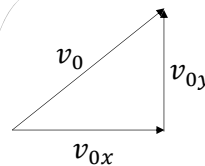
Take going
up as
positive

The **kinetic energy** of the object at time t is

$$\frac{1}{2}mv^2 = \frac{1}{2}m(v_x^2 + v_y^2) = \frac{1}{2}m[v_x^2 + (v_{0y} - gt)^2]$$

$$\text{But } v_x = v_{0x} \rightarrow \frac{1}{2}m(v_x^2 + v_{0y}^2 - 2gv_{0y}t + g^2t^2)$$

$$= \frac{1}{2}m(v_x^2 + v_{0y}^2) - \left(mg \left(v_{0y}t - \frac{1}{2}gt^2 \right) \right) = \frac{1}{2}mv_0^2 - mg(y - y_0)$$



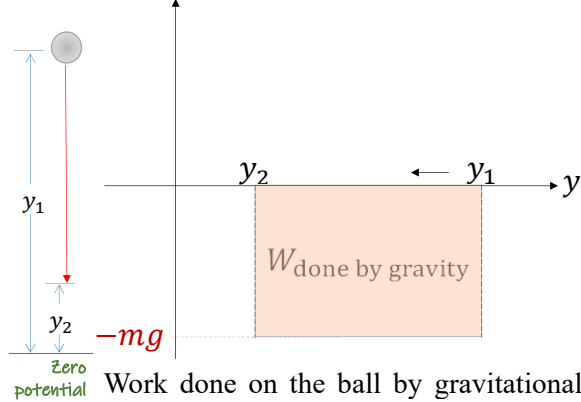
$\rightarrow \left(\frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 \right) + mg(y - y_0) = 0$. Therefore, $\Delta KE + \Delta PE = 0$; **independent** of the details of the trajectory. This means the total mechanical energy is conserved

in an isolated system at any time and at any where, but the form can change between kinetic and potential in whole or in part.

(Law on Conservation of Energy)

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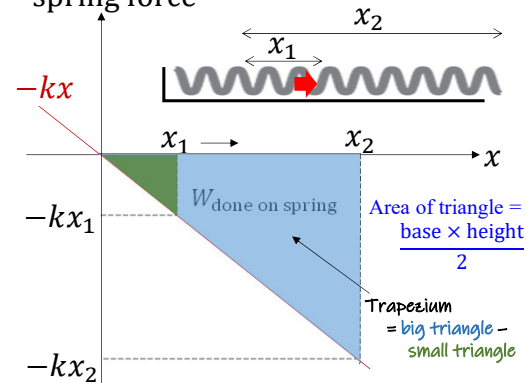
Example 2. Elastic potential energy gravitational force



$$W_{\text{done by gravity}} = (-mg)(y_2 - y_1) = mg(y_1 - y_2) > 0$$

That's why the velocity of the object becomes faster at y_2 .

spring force



Work done on spring by applied (external) force from extension x_1 to extension x_2 :

$$W_{\text{done on spring}} = \frac{(kx_2)(x_2)}{2} - \frac{(kx_1)(x_1)}{2} = \frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2 > 0$$

That's why the spring has stored a lot of elastic potential energy at extension x_2 .

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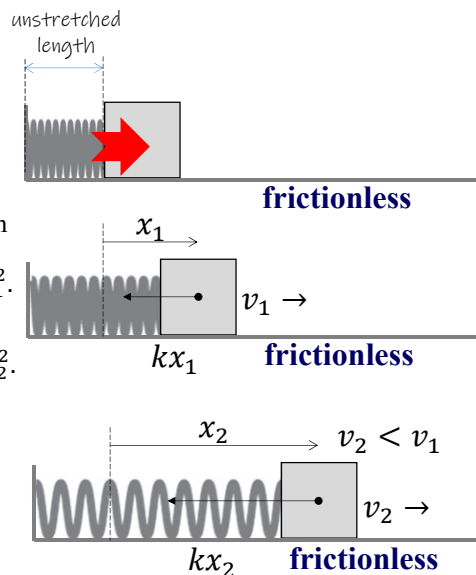
Example 2. Elastic potential energy (cont.)

- Mass of the object is m
- Elastic potential energy stored in spring is $\Delta U_{\text{spring}} = \frac{1}{2}kx^2$, where x is the deformation from the other end (movable) of unstretched length
- At x_1 , **kinetic energy** of the object, $KE_1 = \frac{1}{2}mv_1^2$.
- At x_2 , **kinetic energy** of the object, $KE_2 = \frac{1}{2}mv_2^2$.
- The **change in kinetic energy** of the object,

$$\Delta KE = KE_2 - KE_1 = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = -\left(\frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2\right) < 0$$

So the kinetic energy is given in part (or in whole) to the spring.

That's why the spring has stored a lot of elastic potential energy at x_2 .



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Example 2. Elastic potential energy (cont.)

- The **change in kinetic energy** of the object,

$$\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2 = -\left(\frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2\right) < 0$$

- Elastic potential energy**,

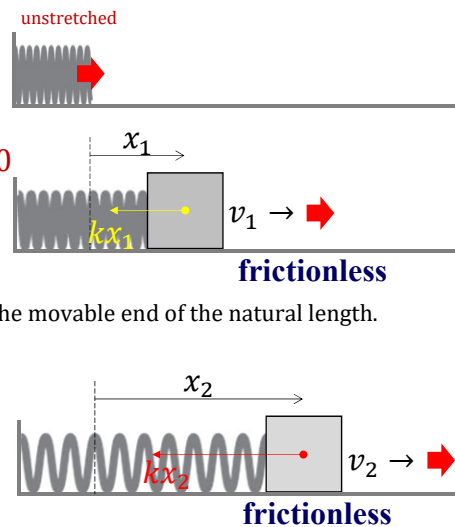
$$U_{\text{spring}} = \frac{1}{2}kx^2, \text{ where } x \text{ is measured from the movable end of the natural length.}$$

- The **change in elastic potential energy**,

$$\Delta U_{\text{spring}} = \frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2 > 0$$

- For an object with a spring attached as shown, change in kinetic energy

$$\Delta KE = W_{\text{done by spring}} = -\Delta U_{\text{spring}}.$$



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Example 2. Elastic potential energy (cont.)

- The position and velocity of an **object with a spring attached** as shown, change in the course of time in such a way that

$$\Delta KE + \Delta U_{\text{spring}} = 0.$$

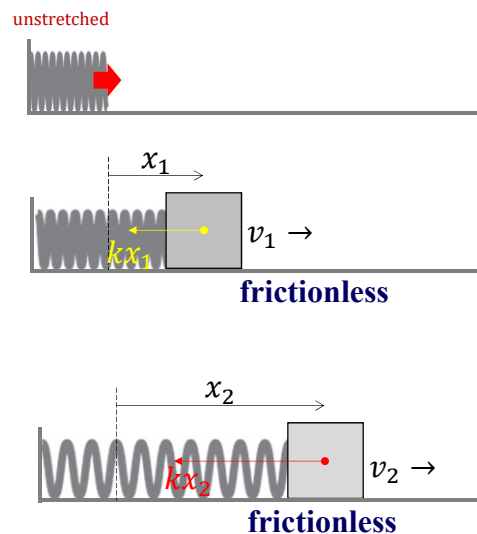
(Law on Conservation of Energy)

Question

What happens when friction acts on the surface in addition to the spring force?

Answer:

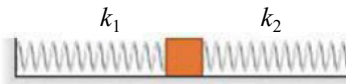
$$\Delta K + \Delta U_{\text{spring}} \neq 0.$$



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Problem. A 3.00-kg block is connected to two ideal horizontal springs having force constants $k_1 = 25.0 \text{ N/cm}$ and $k_2 = 20.0 \text{ N/cm}$ as shown in the figure when the block is in its equilibrium position.

The system is initially in equilibrium on a horizontal, frictionless surface. The block is now pushed 15.0 cm to the right and released from rest.



- (i) What is the maximum speed of the block? Where in the motion does the maximum speed occur?

Answer:

Before the block is released from the right, the left spring is extended 15 cm, and the right spring is compressed 15 cm. The total energy stored in the two springs is $\frac{1}{2}k_1(\Delta x)^2 + \frac{1}{2}k_2(\Delta x)^2$.

The maximum speed occurs when the net acceleration of the block is zero, which happens when the net force is 0, meaning that **the block reaches its equilibrium position.**

The elastic potential energy stored in the two spring will be fully discharged to the block when it is back to the equilibrium position,

$$\frac{1}{2}k_1(\Delta x)^2 + \frac{1}{2}k_2(\Delta x)^2 = \frac{1}{2}mv^2 \rightarrow v = \sqrt{\frac{(k_1 + k_2)(\Delta x)^2}{m}} = \sqrt{\frac{(25 + 20)(15)^2}{3}} = 5.81 \text{ m/s}$$

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- (ii) What is the maximum compression of the spring having force constant k_1 ?



Answer:

Maximum compression happens when the speed (and hence kinetic energy) becomes zero again. By using conservation of mechanical energy again (now taking the final compressed length $\Delta x'$ as the location where spring with constant k_1 is maximally compressed, also meaning that spring with constant k_2 is maximally elongated), and the fact that the kinetic energy comes from the initial compression and extension of length Δx ,

$$\begin{aligned} \cancel{\frac{1}{2}}k_1(\Delta x)^2 + \cancel{\frac{1}{2}}k_2(\Delta x)^2 &= \cancel{\frac{1}{2}}k_1(\Delta x')^2 + \cancel{\frac{1}{2}}k_2(\Delta x')^2 \\ \Rightarrow \cancel{(k_1 + k_2)}(\Delta x)^2 &= \cancel{(k_1 + k_2)}(\Delta x')^2 \Rightarrow (\Delta x)^2 = (\Delta x')^2 \\ \Rightarrow \Delta x' &= \Delta x = 15.0 \text{ cm} \end{aligned}$$

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Example 3. Elastic potential energy and gravitational potential energy

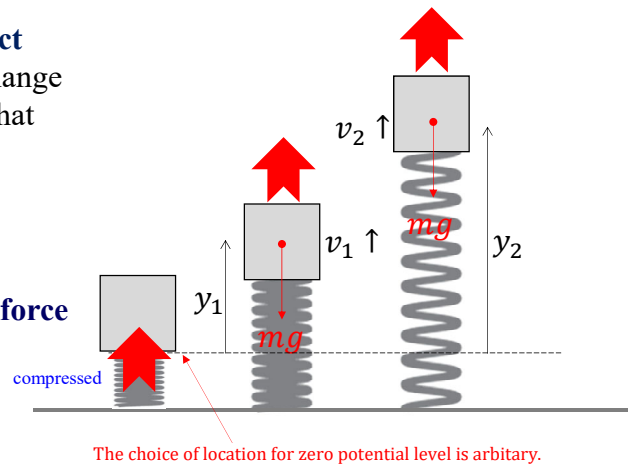
- The position and velocity of an **object with a spring attached** as shown change in the course of time in such a way that

$$\Delta K + \Delta PE + \Delta U_{\text{spring}} = 0.$$

- This is the **conservation of total mechanical energy**.
- The **spring force** and **gravitational force** are examples of **conservative force**.
- Kinetic frictional force** and **fluid resistance force** are examples of **non-conservative force**.

- In the presence of **non-conservative forces**,

$$\Delta K + \Delta PE + \Delta U_{\text{spring}} \neq W_{\text{non-conservative forces}}$$

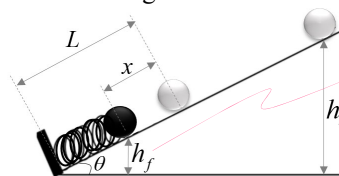


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Problem. A 2 kg mass is at rest when it is 3 m vertically above the ground on a 30° frictionless ramp as shown in the figure. The spring at the bottom of the ramp is 1 m long and has a spring constant of 200 N/m. What is the maximum compression of the spring when the mass collides with it?

Answer:

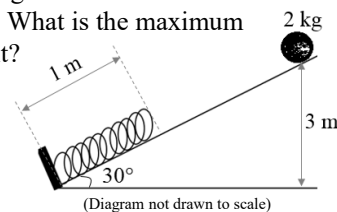
The mass loses all its kinetic energy after sliding down the ramp when the spring is at maximum compression x as shown in the figure below.



By conservation of energy:

$$mgh_i = mgh_f + \frac{1}{2}kx^2$$

$$\Rightarrow h_i = h_f + \frac{kx^2}{2mg}$$



$$h_i = (L - x)\sin\theta + \frac{kx^2}{2mg}$$

$$3 = (1 - x)\sin 30^\circ + \frac{200x^2}{2 \times 2 \times 9.8}$$

$$3 = 0.5 - 0.5x + 5.1x^2$$

$$5.1x^2 - 0.5x - 2.5 = 0$$

$$x = \frac{0.5 \pm \sqrt{(-0.5)^2 - 4(5.1)(-2.5)}}{2 \times 5.1} = \frac{0.5 \pm 7.16}{10.2}$$

$$= 0.75, -0.65$$

Hence the maximum compression is 0.75 m.

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Power

Power is the time rate at which **work** is **done**.

- When a quantity of work ΔW is done during a time interval Δt ,

the **average power**, $P_{\text{average}} = \frac{\Delta W}{\Delta t}$.

- Unit of power, 1 **watt** (W) = 1 J/s = 1 N × 1 m/s.

- (**Instantaneous**) **power**, $P = \frac{dW}{dt}$. $P = Fv$.

$$y = f(v)$$

$$\frac{dy}{dt} = \frac{dy}{dv} \times \frac{dv}{dt}$$

- For a particle, of mass m subject to a force,

$$\begin{aligned} P &= \frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \frac{1}{2} m \left(\frac{d}{dt} (v^2) \right) = \frac{1}{2} m \left(2v \frac{dv}{dt} \right) = m \left(\frac{dv}{dt} \right) v \\ &= (ma)v = Fv \end{aligned}$$

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Problem. A car of mass 1.2×10^3 kg travels along a horizontal road at a speed of 10 m/s. It then accelerates at 0.20 m/s^2 . At the time it begins to accelerate, the total resistive force acting on the car is 160 N.

What is the output power delivered by the car as it begins the acceleration?

Answer:

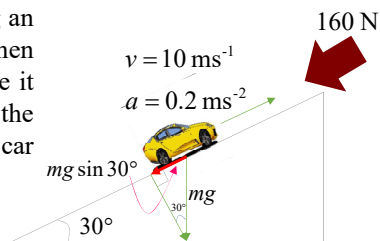
$$\begin{aligned} \text{Power} &= \frac{\text{Work done by Engine}}{\text{Time}} = \frac{(\text{Thrust of Engine}) \times \text{Distance}}{\text{Time}} \\ &= \text{Thrust of Engine} \times \frac{\text{Distance}}{\text{Time}} = \text{Thrust of Engine} \times \text{Velocity} \end{aligned}$$

But the thrust of engine has to overcome the total resistive force and to produce the acceleration.

$$\begin{aligned} \text{So Power of the car} &= [160 + ma] \times 10 \text{ m/s} \\ &= [160 + (1.2 \times 10^3) \times 0.20] \times 10 \\ &= 4.0 \text{ kW} \end{aligned}$$

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Problem. Another car of mass 1.2×10^3 kg travels along an incline plane as shown at a speed of 10 m/s. It then accelerates at 0.20 ms^{-2} up the incline plane. At the time it begins to accelerate, the total resistive force acting on the car is 160 N. What is the output power delivered by the car as it begins the acceleration?



Answer:

$$\text{Power} = \frac{\text{Work done by Engine}}{\text{Time}} = \frac{(\text{Thrust of Engine}) \times \text{Distance}}{\text{Time}}$$

$$= \text{Thrust of Engine} \times \frac{\text{Distance}}{\text{Time}} = \text{Thrust of Engine} \times \text{Velocity}$$

The thrust of engine now has to overcome the total resistive force, and also to carry up the **burden** (projected weight along the incline plane), and to produce the acceleration.

$$\begin{aligned} \text{Power} &= \text{Thrust of Engine} \times \text{Velocity} \\ &= (\text{resistive force} + mg \sin \theta + ma) \times 10 \text{ m/s} \\ &= (160 \text{ N} + 1.2 \times 10^3 \times 9.8 \times 0.5 \text{ N} + 1.2 \times 10^3 \times 0.2 \text{ N}) \times 10 \text{ m/s} \\ &= (160 \text{ N} + 5880 \text{ N} + 240 \text{ N}) \times 10 \text{ m/s} = 62.8 \text{ kW} \end{aligned}$$

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Problem. A 650-kg elevator starts from rest. It moves upward for 3.00 s with constant acceleration until it reaches its cruising speed of 1.75 m/s.

(i) What is the average power of the elevator motor during this period?

Answer:

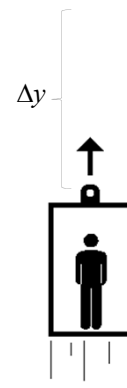
The distance moved upward in the first 3.00 s is

$$\Delta y = \bar{v}t = \left(\frac{0 \text{ m/s} + 1.75 \text{ m/s}}{2} \right) (3.00 \text{ s}) = 2.63 \text{ m}$$

Work done by the lift motor is given to increase the speed of the lift, and, to elevate the lift for a height of 2.63 m.

$$\begin{aligned} W_{\text{done by lift motor}} &= \frac{1}{2}mv_f^2 + mgh \\ &= \frac{1}{2}(650 \text{ kg})(1.75 \text{ m/s})^2 + (650 \text{ kg})(9.8 \text{ m/s}^2)(2.63 \text{ m}) = 1.77 \times 10^4 \text{ J} \end{aligned}$$

$$\bar{P} = \frac{W_{\text{done by lift motor}}}{t} = \frac{1.77 \times 10^4 \text{ J}}{3.00 \text{ s}} = \boxed{5.92 \times 10^3 \text{ W}}$$



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$$\bar{P} = \frac{W_{\text{done by lift motor}}}{t} = \frac{1.77 \times 10^4 \text{ J}}{3.00 \text{ s}} = \boxed{5.92 \times 10^3 \text{ W}}$$

starts from rest for 3.00 s with constant upward acceleration

(ii) How does this power compare with the motor power when the elevator moves at its cruising speed?

Answer:

When moving upward at **constant speed** ($v = 1.75 \text{ m/s}$) the applied force of the lift motor only need to overcome the weight.

$$P = F \times v = (mg) \times v$$

$$= (650 \text{ kg})(9.80 \text{ m/s}^2)(1.75 \text{ m/s}) = \boxed{1.11 \times 10^4 \text{ W}}$$

Because Power also depends on the magnitude of the constant velocity.

$$5.92 \times 10^3 \text{ W}$$

Why larger?

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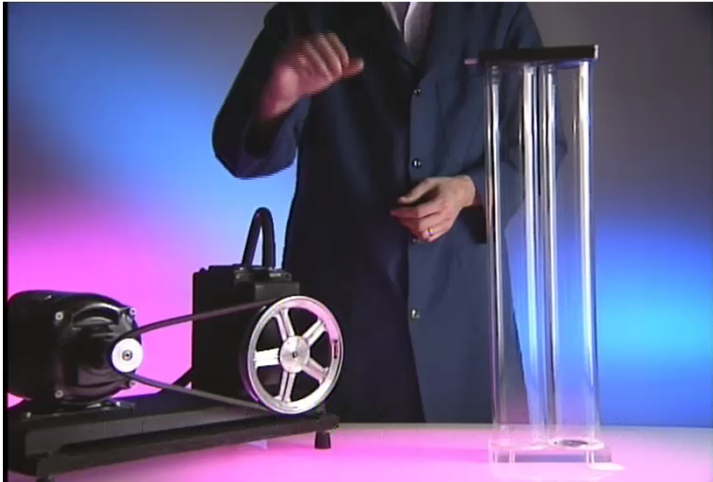
The Drag Force of Air



What if the air resistance is removed?

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The Drag Force of Air



How do we harness power from the drag force of air?

By the use of windmill!!

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Problem. A windmill turns in response to a force of high-speed air resistance $R = \frac{1}{2} D \rho A v^2$, where v is the wind speed, A is the area of the circular face of the windmill, ρ is the density of air from the front, and D is the drag coefficient.

For a home windmill with radius $r = 1.50$ m for the circular face, calculate the power available if the wind speed is 8.00 m/s. If the power delivered to the generator is limited by the efficiency of the system at about 25%, discuss whether the home windmill can meet the need of a typical home for 3 kW of electric power.

($\rho = 1.20$ kg/m³, and $D = 1$)

Answer:

$$\text{Power} = \frac{\text{Energy}}{\text{Time}} = \frac{\text{Force} \times \text{distance}}{\text{Time}} = \text{Force} \times \left(\frac{\text{distance}}{\text{Time}} \right) = \text{Force} \times \text{velocity}$$

$$\text{Perfect power from the windmill system} = R \times v = \frac{1}{2} D \rho A v^2 \times v = \frac{1}{2} D \rho (\pi r^2) v^3$$

$$\text{Actual power from the windmill system} = \frac{1}{2} D \rho \pi r^2 v^3 \times 25\% = \frac{1}{8} D \rho \pi r^2 v^3$$

$$= \frac{1}{8} \times 1 \times (1.20 \text{ kg/m}^3) \pi (1.5 \text{ m})^2 (8 \text{ m/s})^3 = 0.5429 \times 10^3 \text{ W} = 0.5429 \text{ kW} < 3 \text{ kW} !!$$

But the home owner can use a windfarm of 6 windmills to supply the power to the home if space is available.

(Alternate method is to adopt solar panels.)



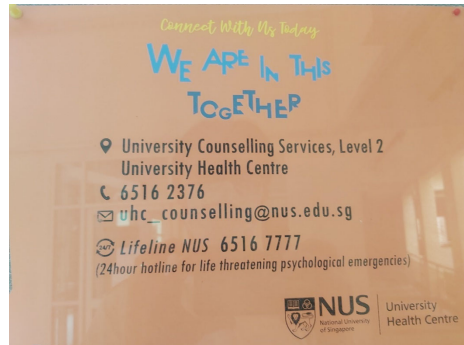
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Helplines to de-stress

We are not superman/supergirl. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
 Institute of Mental Health's Helpline: 6389 2222
 Singapore Association of Mental Health Helpline: 1800 283 7019